

Task 1: Sparse Networks from Scratch

In the first task, students will train sparse networks with a fixed random connectivity mask. The mask M is generated at initialization and remains unchanged throughout training. Students should compare at least two optimization methods: SGD with momentum and Adam. The goal is to investigate how different optimizers behave when training highly sparse networks. Suggested questions include:

- How does sparsity affect convergence speed?
- How does accuracy degrade as sparsity increases?
- Do different optimizers behave differently under high sparsity?

```
# from google.colab import drive
# drive.mount('/content/drive')
```

Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/")

Load libraries

```
import os
import sys
import torch
import torch.nn as nn
import torch.optim as optim
import joblib
import torchvision
import torchvision.transforms as transforms

import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd
```

```
# quarto preview 01_task_1.ipynb --to pdf
```

```
# model_path = "/content/drive/MyDrive/Posgrados/2025/Duke/ECE 685D/Final project"  
# sys.path.append(model_path)
```

```
model_path = "."
```

```
from models import ResNetCIFAR
```

```
use_pin_memory = torch.cuda.is_available()
```

```
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
```

```
num_workers = min(2, os.cpu_count())
```

Load data

```
# Standard CIFAR-10 transforms  
transform = transforms.Compose(  
    [  
        transforms.ToTensor(),  
        transforms.Normalize((0.4914, 0.4822, 0.4465), (0.2023, 0.1994, 0.2010)),  
    ]  
)
```

```
trainset = torchvision.datasets.CIFAR10(  
    root=f"{model_path}/data", train=True, download=True, transform=transform  
)
```

```
trainloader = torch.utils.data.DataLoader(  
    trainset,  
    batch_size=128,  
    shuffle=True,  
    num_workers=num_workers,  
    pin_memory=use_pin_memory,  
)
```

```
testset = torchvision.datasets.CIFAR10(  
    root=f"{model_path}/data", train=False, download=True, transform=transform  
)
```

```
testloader = torch.utils.data.DataLoader(
    testset,
    batch_size=128,
    shuffle=False,
    num_workers=num_workers,
    pin_memory=use_pin_memory,
)
```

Configuration

```
sparsities = [0.90, 0.95, 0.98]
optimizers = ["sgd", "adam"]
```

Main Experiment

```
epochs = 15
criterion = nn.CrossEntropyLoss()
```

```
output_dir = "/content/drive/MyDrive/Posgrads/2025/Duke/ECE 685D/Final project/results"
```

Helper Functions

```
def get_static_masks(model, sparsity):
    """Generates the fixed binary mask M"""
    masks = {}
    for name, param in model.named_parameters():
        if "conv" in name or "linear" in name:
            mask = torch.rand(param.size()).to(device)
            mask = (mask > sparsity).float()
            masks[name] = mask
        else:
            masks[name] = torch.ones(param.size()).to(device)
    return masks

@torch.no_grad()
```

```
def apply_masks(model, masks):
    """Applies  $W = M * \theta$ """
    for name, param in model.named_parameters():
        param.mul_(masks[name])
```

Actual experiment

This section runs the full experimental loop over sparsity levels and optimizers. For each configuration, a model is initialized, a fixed random mask is applied, and training proceeds while enforcing the same sparse connectivity throughout. Performance metrics are tracked at every epoch and saved for later analysis.

```
for s in sparsities:
    for opt_name in optimizers:
        print(f"\nStarting Experiment: Sparsity={s*100}%, Optimizer={opt_name}")

        # Initialize
        model = ResNetCIFAR().to(device)
        masks = get_static_masks(model, s)
        apply_masks(model, masks)

        if opt_name == "sgd":
            optimizer = optim.SGD(
                model.parameters(), lr=0.1, momentum=0.9, weight_decay=5e-4
            )
        else:
            optimizer = optim.Adam(model.parameters(), lr=0.001, weight_decay=5e-4)

        history = {"train_loss": [], "train_acc": [], "test_loss": [], "test_acc": []}

        for epoch in range(epochs):
            # --- Training Phase ---
            model.train()
            train_loss, train_correct, train_total = 0.0, 0, 0

            for inputs, labels in trainloader:
                inputs, labels = inputs.to(device), labels.to(device)
                optimizer.zero_grad()
                outputs = model(inputs)
                loss = criterion(outputs, labels)
                loss.backward()
```

```

optimizer.step()
apply_masks(model, masks) # Keep connectivity fixed

train_loss += loss.item()
_, predicted = outputs.max(1)
train_total += labels.size(0)
train_correct += predicted.eq(labels).sum().item()

# --- Evaluation Phase ---
model.eval()
test_loss, test_correct, test_total = 0.0, 0, 0
with torch.no_grad():
    for inputs, labels in testloader:
        inputs, labels = inputs.to(device), labels.to(device)
        outputs = model(inputs)
        loss = criterion(outputs, labels)
        test_loss += loss.item()
        _, predicted = outputs.max(1)
        test_total += labels.size(0)
        test_correct += predicted.eq(labels).sum().item()

# Record Metrics
history["train_loss"].append(train_loss / len(trainloader))
history["train_acc"].append(100.0 * train_correct / train_total)
history["test_loss"].append(test_loss / len(testloader))
history["test_acc"].append(100.0 * test_correct / test_total)

print(
    f"Epoch {epoch+1}: Train Loss {history['train_loss'][-1]:.4f} | Test Acc {hi
)

# --- Save Joblib ---
experiment_results = {
    "sparsity": s,
    "optimizer": opt_name,
    "train_loss": history["train_loss"],
    "train_acc": history["train_acc"],
    "test_loss": history["test_loss"],
    "test_acc": history["test_acc"],
    "masks": masks,
    "model_state": model.state_dict(),
}

```

```

filename = os.path.join(
    output_dir, f"task_1_{opt_name}_{int(s*100)}_sparsity.joblib"
)
joblib.dump(experiment_results, filename)
print(f"Results saved to {filename}!")

```

Starting Experiment: Sparsity=90.0%, Optimizer=sgd

```

Epoch 1: Train Loss 1.4800 | Test Acc 46.15%
Epoch 2: Train Loss 1.0508 | Test Acc 57.55%
Epoch 3: Train Loss 0.8884 | Test Acc 54.54%
Epoch 4: Train Loss 0.8074 | Test Acc 51.62%
Epoch 5: Train Loss 0.7596 | Test Acc 69.40%
Epoch 6: Train Loss 0.7299 | Test Acc 64.62%
Epoch 7: Train Loss 0.7096 | Test Acc 67.02%
Epoch 8: Train Loss 0.6902 | Test Acc 68.87%
Epoch 9: Train Loss 0.6775 | Test Acc 70.44%
Epoch 10: Train Loss 0.6648 | Test Acc 69.63%
Epoch 11: Train Loss 0.6513 | Test Acc 70.86%
Epoch 12: Train Loss 0.6478 | Test Acc 60.67%
Epoch 13: Train Loss 0.6390 | Test Acc 70.60%
Epoch 14: Train Loss 0.6302 | Test Acc 71.90%
Epoch 15: Train Loss 0.6326 | Test Acc 70.14%

```

Results saved to /content/drive/MyDrive/Posgrados/2025/Duke/ECE 685D/Final project/results/t

Starting Experiment: Sparsity=90.0%, Optimizer=adam

```

Epoch 1: Train Loss 1.5871 | Test Acc 48.50%
Epoch 2: Train Loss 1.1645 | Test Acc 58.19%
Epoch 3: Train Loss 1.0149 | Test Acc 60.58%
Epoch 4: Train Loss 0.9248 | Test Acc 66.05%
Epoch 5: Train Loss 0.8605 | Test Acc 64.21%
Epoch 6: Train Loss 0.8024 | Test Acc 67.34%
Epoch 7: Train Loss 0.7640 | Test Acc 68.56%
Epoch 8: Train Loss 0.7243 | Test Acc 70.47%
Epoch 9: Train Loss 0.6920 | Test Acc 70.43%
Epoch 10: Train Loss 0.6628 | Test Acc 70.66%
Epoch 11: Train Loss 0.6403 | Test Acc 71.42%
Epoch 12: Train Loss 0.6167 | Test Acc 71.68%
Epoch 13: Train Loss 0.5946 | Test Acc 73.01%
Epoch 14: Train Loss 0.5739 | Test Acc 72.12%
Epoch 15: Train Loss 0.5620 | Test Acc 73.34%

```

Results saved to /content/drive/MyDrive/Posgradados/2025/Duke/ECE 685D/Final project/results/t

Starting Experiment: Sparsity=95.0%, Optimizer=sgd

Epoch 1:	Train Loss	1.6498		Test Acc	38.35%
Epoch 2:	Train Loss	1.2002		Test Acc	57.12%
Epoch 3:	Train Loss	1.0380		Test Acc	59.63%
Epoch 4:	Train Loss	0.9561		Test Acc	60.00%
Epoch 5:	Train Loss	0.9054		Test Acc	52.98%
Epoch 6:	Train Loss	0.8657		Test Acc	65.12%
Epoch 7:	Train Loss	0.8465		Test Acc	62.64%
Epoch 8:	Train Loss	0.8222		Test Acc	63.60%
Epoch 9:	Train Loss	0.8056		Test Acc	65.55%
Epoch 10:	Train Loss	0.7962		Test Acc	67.65%
Epoch 11:	Train Loss	0.7894		Test Acc	66.20%
Epoch 12:	Train Loss	0.7805		Test Acc	55.36%
Epoch 13:	Train Loss	0.7666		Test Acc	60.81%
Epoch 14:	Train Loss	0.7665		Test Acc	67.60%
Epoch 15:	Train Loss	0.7544		Test Acc	61.46%

Results saved to /content/drive/MyDrive/Posgradados/2025/Duke/ECE 685D/Final project/results/t

Starting Experiment: Sparsity=95.0%, Optimizer=adam

Epoch 1:	Train Loss	1.8323		Test Acc	36.19%
Epoch 2:	Train Loss	1.5417		Test Acc	41.56%
Epoch 3:	Train Loss	1.4106		Test Acc	42.55%
Epoch 4:	Train Loss	1.3250		Test Acc	45.08%
Epoch 5:	Train Loss	1.2622		Test Acc	48.91%
Epoch 6:	Train Loss	1.2182		Test Acc	49.05%
Epoch 7:	Train Loss	1.1826		Test Acc	50.83%
Epoch 8:	Train Loss	1.1545		Test Acc	54.83%
Epoch 9:	Train Loss	1.1285		Test Acc	51.85%
Epoch 10:	Train Loss	1.1068		Test Acc	56.14%
Epoch 11:	Train Loss	1.0861		Test Acc	56.15%
Epoch 12:	Train Loss	1.0698		Test Acc	57.18%
Epoch 13:	Train Loss	1.0574		Test Acc	55.74%
Epoch 14:	Train Loss	1.0411		Test Acc	57.95%
Epoch 15:	Train Loss	1.0289		Test Acc	57.92%

Results saved to /content/drive/MyDrive/Posgradados/2025/Duke/ECE 685D/Final project/results/t

Starting Experiment: Sparsity=98.0%, Optimizer=sgd

Epoch 1:	Train Loss	1.7946		Test Acc	36.59%
Epoch 2:	Train Loss	1.5153		Test Acc	46.35%
Epoch 3:	Train Loss	1.4071		Test Acc	45.13%
Epoch 4:	Train Loss	1.3515		Test Acc	42.58%

```
Epoch 5: Train Loss 1.3171 | Test Acc 45.01%
Epoch 6: Train Loss 1.2975 | Test Acc 42.05%
Epoch 7: Train Loss 1.2746 | Test Acc 47.86%
Epoch 8: Train Loss 1.2583 | Test Acc 50.42%
Epoch 9: Train Loss 1.2576 | Test Acc 44.14%
Epoch 10: Train Loss 1.2435 | Test Acc 50.08%
Epoch 11: Train Loss 1.2302 | Test Acc 46.97%
Epoch 12: Train Loss 1.2275 | Test Acc 46.45%
Epoch 13: Train Loss 1.2237 | Test Acc 44.64%
Epoch 14: Train Loss 1.2136 | Test Acc 36.23%
Epoch 15: Train Loss 1.2156 | Test Acc 50.50%
Results saved to /content/drive/MyDrive/Posgrados/2025/Duke/ECE 685D/Final project/results/t
```

```
Starting Experiment: Sparsity=98.0%, Optimizer=adam
Epoch 1: Train Loss 1.9828 | Test Acc 30.40%
Epoch 2: Train Loss 1.7337 | Test Acc 37.47%
Epoch 3: Train Loss 1.6010 | Test Acc 40.73%
Epoch 4: Train Loss 1.5292 | Test Acc 41.95%
Epoch 5: Train Loss 1.4710 | Test Acc 43.88%
Epoch 6: Train Loss 1.4234 | Test Acc 45.62%
Epoch 7: Train Loss 1.3807 | Test Acc 45.25%
Epoch 8: Train Loss 1.3495 | Test Acc 45.60%
Epoch 9: Train Loss 1.3253 | Test Acc 46.64%
Epoch 10: Train Loss 1.3034 | Test Acc 47.59%
Epoch 11: Train Loss 1.2862 | Test Acc 48.19%
Epoch 12: Train Loss 1.2712 | Test Acc 49.35%
Epoch 13: Train Loss 1.2562 | Test Acc 48.64%
Epoch 14: Train Loss 1.2407 | Test Acc 46.10%
Epoch 15: Train Loss 1.2298 | Test Acc 47.28%
Results saved to /content/drive/MyDrive/Posgrados/2025/Duke/ECE 685D/Final project/results/t
```

The printed outputs provide a quick view of training loss and test accuracy across epochs for each configuration. As sparsity increases, performance generally degrades, reflecting the reduced capacity of the network. Differences between optimizers also become more pronounced at higher sparsities, with Adam often converging more smoothly than SGD.

These logs only show a subset of the recorded metrics. Additional information such as training accuracy, test loss, and the full training history are stored in the saved joblib files for more detailed analysis.

Evaluation

Functions

```
def plot_training_and_validation(results):
    sns.set_theme(style="whitegrid", context="notebook", palette="deep")

    sparsities = [0.90, 0.95, 0.98]
    opts = ["sgd", "adam"]

    for s in sparsities:
        rows = []

        # --- Reshape data into tidy format ---
        for opt in opts:
            if s in results[opt]:
                n_epochs = len(results[opt][s]["train_loss"])
                for epoch in range(n_epochs):
                    rows.append(
                        {
                            "epoch": epoch,
                            "optimizer": opt.upper(),
                            "split": "Train",
                            "loss": results[opt][s]["train_loss"][epoch],
                            "accuracy": results[opt][s]["train_acc"][epoch],
                        }
                    )
                rows.append(
                    {
                        "epoch": epoch,
                        "optimizer": opt.upper(),
                        "split": "Test",
                        "loss": results[opt][s]["test_loss"][epoch],
                        "accuracy": results[opt][s]["test_acc"][epoch],
                    }
                )

        df = pd.DataFrame(rows)

        # --- Create 2x2 figure ---
        fig, axes = plt.subplots(2, 2, figsize=(14, 10), sharex=True)
```

```

fig.suptitle(
    f"Optimization Analysis at {int(s*100)}% Sparsity",
    fontsize=18,
    weight="bold",
)

# --- Train Loss ---
sns.lineplot(
    data=df[df["split"] == "Train"],
    x="epoch",
    y="loss",
    hue="optimizer",
    linewidth=2.5,
    ax=axes[0, 0],
)
axes[0, 0].set_title("Train Loss")
axes[0, 0].set_ylabel("Loss")
axes[0, 0].legend(title="")

# --- Test Loss ---
sns.lineplot(
    data=df[df["split"] == "Test"],
    x="epoch",
    y="loss",
    hue="optimizer",
    linewidth=2.5,
    ax=axes[0, 1],
)
axes[0, 1].set_title("Test Loss")
axes[0, 1].set_ylabel("Loss")
axes[0, 1].legend(title="")

# --- Train Accuracy ---
sns.lineplot(
    data=df[df["split"] == "Train"],
    x="epoch",
    y="accuracy",
    hue="optimizer",
    linewidth=2.5,
    ax=axes[1, 0],
)
axes[1, 0].set_title("Train Accuracy")

```

```

axes[1, 0].set_xlabel("Epoch")
axes[1, 0].set_ylabel("Accuracy (%)")
axes[1, 0].legend(title="")

# --- Test Accuracy ---
sns.lineplot(
    data=df[df["split"] == "Test"],
    x="epoch",
    y="accuracy",
    hue="optimizer",
    linewidth=2.5,
    ax=axes[1, 1],
)
axes[1, 1].set_title(
    "Test Accuracy",
)
axes[1, 1].set_xlabel("Epoch")
axes[1, 1].set_ylabel("Accuracy (%)")
axes[1, 1].legend(title="")

# --- Final polish ---
for ax in axes.flat:
    ax.grid(True, alpha=0.3)

sns.despine()
plt.tight_layout()

plt.savefig(
    f"img/task_1_sparsity_{int(s*100)}.png", dpi=300, bbox_inches="tight"
)
plt.show()

```

```

def plot_accuracy_degradation(results):
    sns.set_theme(style="whitegrid", context="paper", palette="deep")

    sparsities = [0.90, 0.95, 0.98]
    opts = ["sgd", "adam"]

    # --- Build tidy dataframe ---
    rows = []
    for opt in opts:
        for s in sparsities:

```

```

        if s in results[opt]:
            rows.append(
                {
                    "optimizer": opt.upper(),
                    "sparsity": int(s * 100),
                    "final_acc": results[opt][s]["test_acc"][-1],
                }
            )

df = pd.DataFrame(rows)

fig, ax = plt.subplots(figsize=(5, 3))

sns.lineplot(
    data=df,
    x="sparsity",
    y="final_acc",
    hue="optimizer",
    marker="o",
    linewidth=2.5,
    ax=ax,
)

# --- ADD VALUE LABELS ---
for _, row in df.iterrows():
    ax.text(
        row["sparsity"],
        row["final_acc"],
        f"{row['final_acc']:.1f}",
        ha="center",
        va="bottom",
        fontsize=8,
    )

ax.set_title(
    "Impact of Sparsity on Final Test Accuracy",
    weight="bold",
)
ax.set_xlabel("Sparsity Level (%)")
ax.set_ylabel("Final Test Accuracy (%)")

ax.set_xticks([90, 95, 98])

```

```

ax.legend(title="")

ax.grid(True, alpha=0.3)
sns.despine()

plt.tight_layout()
plt.savefig("img/task_1_accuracy_vs_sparsity.png", dpi=300, bbox_inches="tight")
plt.show()

```

```

def report_layer_sparsity(results, opt="sgd", s=0.98):
    """
    Quantifies the sparsity level of different network components.
    Formula: sparsity(W) = 1 - (||W||_0 / |W|)
    """
    if opt in results and s in results[opt]:
        masks = results[opt][s]["masks"]

        print(f"--- Sparsity Statistics: {opt.upper()} | {int(s*100)}% Target ---")
        print(f"{'Layer Name':<35} | {'Active Weights':<15} | {'Actual Sparsity':<15}")
        print("-" * 70)

        total_params = 0
        total_nonzero = 0

        for name, mask in masks.items():
            # Calculate metrics using the project formula
            n_params = mask.numel()
            n_nonzero = torch.count_nonzero(mask).item()

            # Project formula: 1 - (nonzero / total)
            layer_sparsity = 1.0 - (n_nonzero / n_params)

            # Accumulate for global stats
            total_params += n_params
            total_nonzero += n_nonzero

            print(f"{name:<35} | {n_nonzero:<15} | {layer_sparsity:.4f}")

        overall_sparsity = 1.0 - (total_nonzero / total_params)
        print("-" * 70)
        print(
            f"{'TOTAL NETWORK SPARSITY':<35} | {total_nonzero:<15} | {overall_sparsity:.4f}\n"

```

```

    )
else:
    print(f"Data for {opt} at {s} sparsity not found in results.")

def visualize_kernels(results, opt="sgd", s=0.90, layer_name="layer1.0.conv1.weight"):
    sns.set_theme(style="white", context="notebook")

    if opt in results and s in results[opt]:

        weights = results[opt][s]["model_state"][layer_name]

        kernels = weights[:,16, 0, :, :].detach().cpu().numpy()

        fig, axes = plt.subplots(4, 4, figsize=(6, 6))

        fig.suptitle(
            f"Kernels ({layer_name})\n{opt.upper()} | {int(s*100)}% Sparsity",
            fontsize=14,
            weight="bold",
        )

        ims = []

        for i, ax in enumerate(axes.flat):
            im = ax.imshow(kernels[i], cmap="viridis", interpolation="nearest")
            ims.append(im)

            ax.set_title(f"F{i+1}", fontsize=8)
            ax.axis("off")

        # --- shared colorbar (makes comparison meaningful) ---
        cbar = fig.colorbar(ims[0], ax=axes, fraction=0.025, pad=0.02)
        cbar.ax.tick_params(labelsize=8)

        plt.tight_layout()
        plt.show()

    else:
        print(f"Data for {opt} at {s} sparsity not found.")

```

Read results

```
output_dir = "results"
```

```
# Patch torch.load to force CPU
_original_torch_load = torch.load
```

```
def cpu_torch_load(*args, **kwargs):
    kwargs["map_location"] = torch.device("cpu")
    return _original_torch_load(*args, **kwargs)
```

```
torch.load = cpu_torch_load
```

```
results = {}
```

```
for opt_name in optimizers:
    results[opt_name] = {}
    for s in sparsities:
        file_path = os.path.join(
            output_dir, f"task_1_{opt_name}_{int(s*100)}_sparsity.joblib"
        )

        if os.path.exists(file_path):
            data = joblib.load(file_path)
            results[opt_name][s] = data
            print(f"Loaded: {file_path}")
        else:
            print(f"Missing: {file_path}")
```

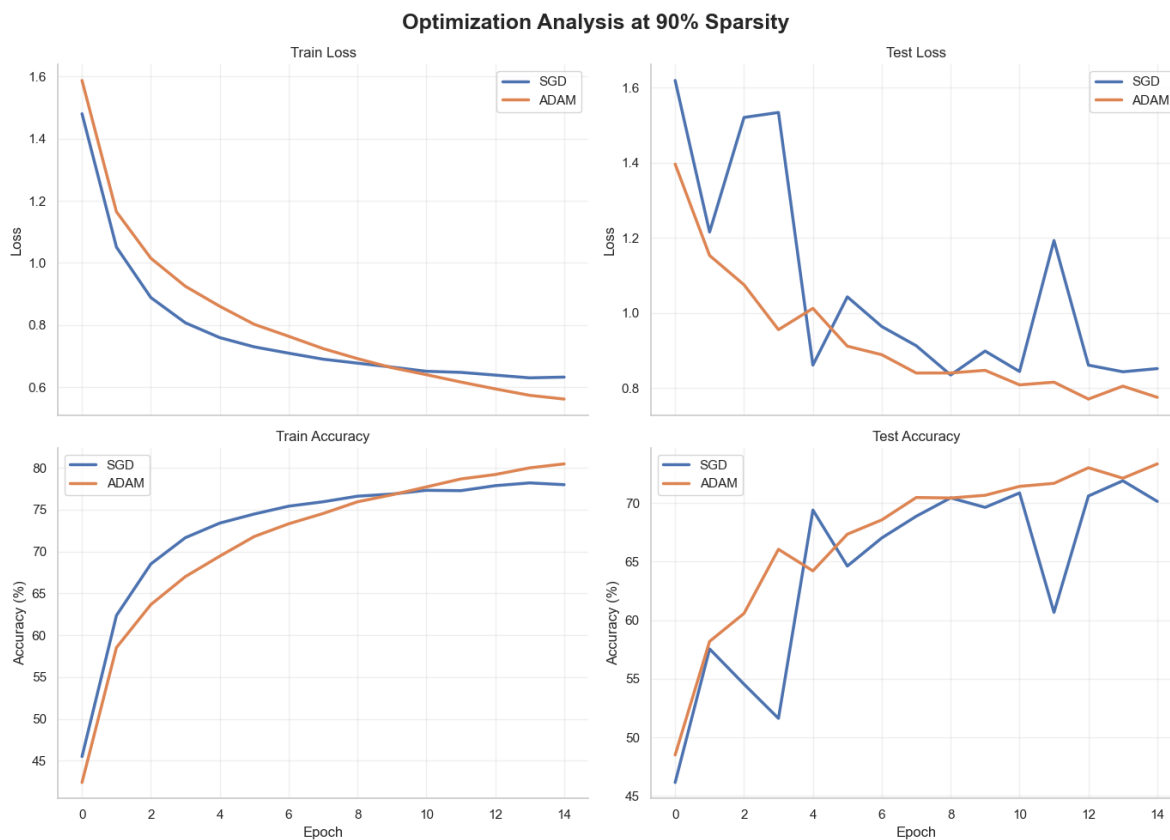
```
Loaded: results/task_1_sgd_90_sparsity.joblib
Loaded: results/task_1_sgd_95_sparsity.joblib
Loaded: results/task_1_sgd_98_sparsity.joblib
Loaded: results/task_1_adam_90_sparsity.joblib
Loaded: results/task_1_adam_95_sparsity.joblib
Loaded: results/task_1_adam_98_sparsity.joblib
```

Analysis

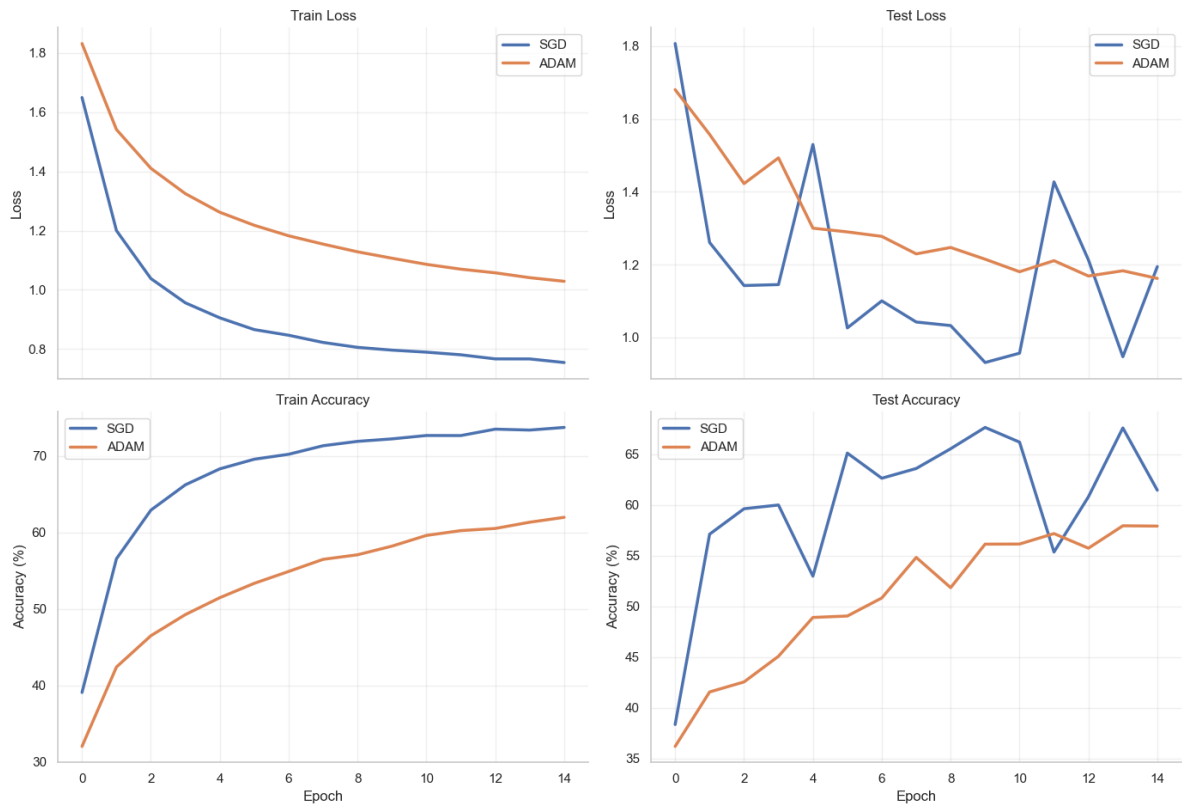
Training and Validation Curves

- How does sparsity affect convergence speed?

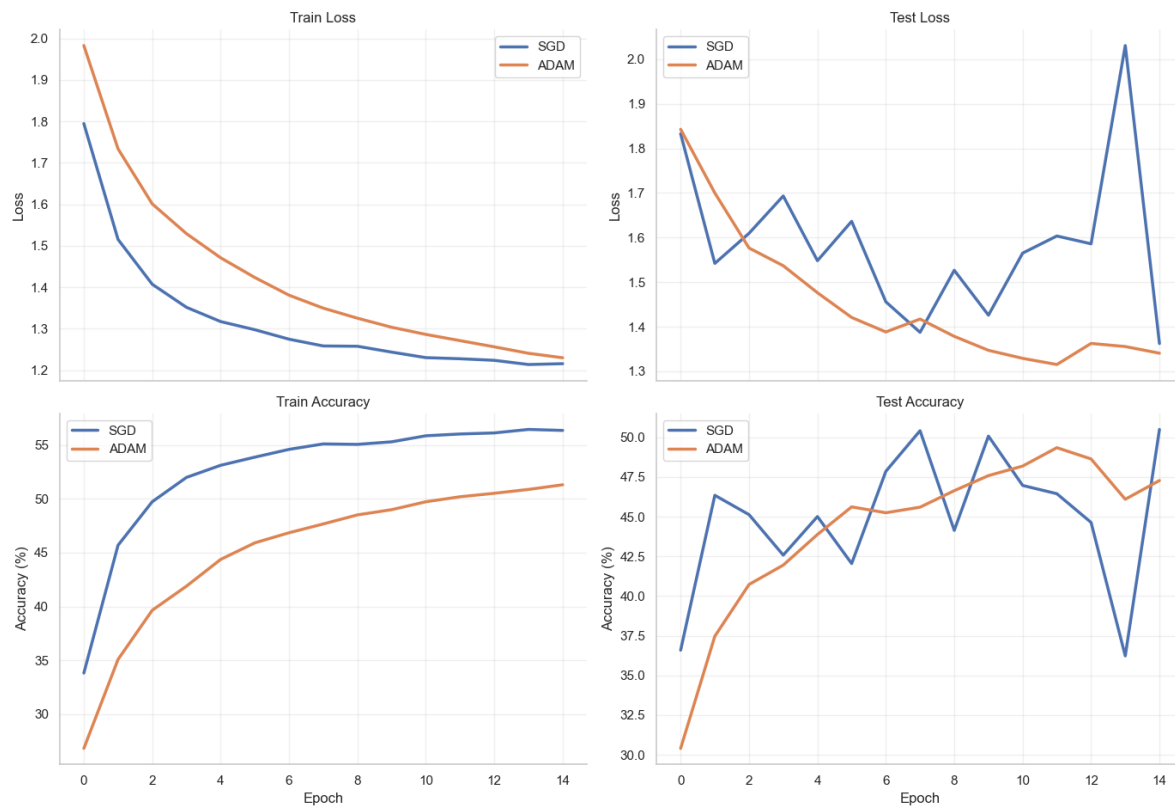
```
plot_training_and_validation(results)
```



Optimization Analysis at 95% Sparsity



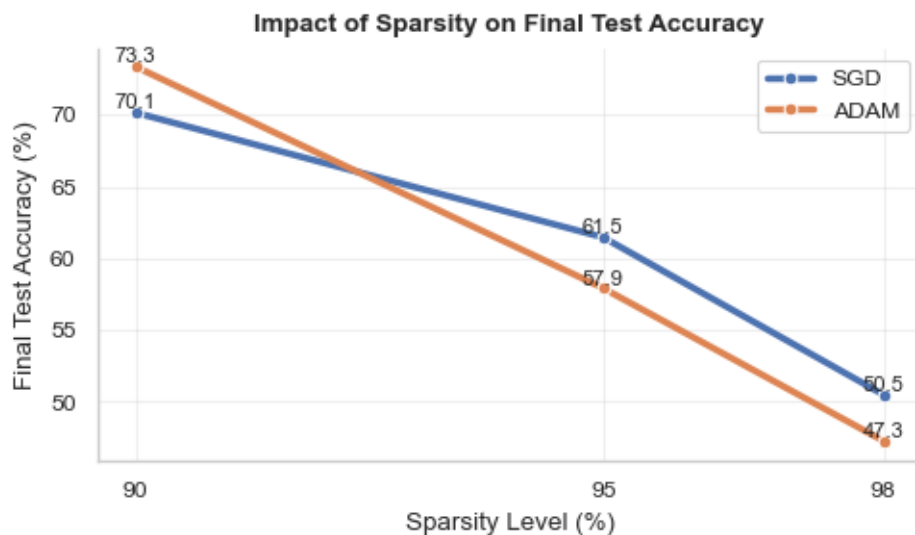
Optimization Analysis at 98% Sparsity



Accuracy vs. Sparsity (Degradation)

- How does accuracy degrade as sparsity increases?

```
plot_accuracy_degradation(results)
```



Layer sparsity

This is from the evaluation metrics

```
for opt in optimizers:
    for s in sparsities:
        report_layer_sparsity(results, opt=opt, s=s)
```

--- Sparsity Statistics: SGD | 90% Target ---

Layer Name	Active Weights	Actual Sparsity

conv1.weight	53	0.8773
bn1.weight	16	0.0000
bn1.bias	16	0.0000
layer1.0.conv1.weight	229	0.9006
layer1.0.bn1.weight	16	0.0000
layer1.0.bn1.bias	16	0.0000
layer1.0.conv2.weight	225	0.9023
layer1.0.bn2.weight	16	0.0000
layer1.0.bn2.bias	16	0.0000
layer1.1.conv1.weight	240	0.8958
layer1.1.bn1.weight	16	0.0000
layer1.1.bn1.bias	16	0.0000
layer1.1.conv2.weight	242	0.8950
layer1.1.bn2.weight	16	0.0000

layer1.1.bn2.bias	16	0.0000
layer1.2.conv1.weight	219	0.9049
layer1.2.bn1.weight	16	0.0000
layer1.2.bn1.bias	16	0.0000
layer1.2.conv2.weight	257	0.8885
layer1.2.bn2.weight	16	0.0000
layer1.2.bn2.bias	16	0.0000
layer2.0.conv1.weight	502	0.8911
layer2.0.bn1.weight	32	0.0000
layer2.0.bn1.bias	32	0.0000
layer2.0.conv2.weight	948	0.8971
layer2.0.bn2.weight	32	0.0000
layer2.0.bn2.bias	32	0.0000
layer2.0.shortcut.0.weight	512	0.0000
layer2.0.shortcut.1.weight	32	0.0000
layer2.0.shortcut.1.bias	32	0.0000
layer2.1.conv1.weight	902	0.9021
layer2.1.bn1.weight	32	0.0000
layer2.1.bn1.bias	32	0.0000
layer2.1.conv2.weight	891	0.9033
layer2.1.bn2.weight	32	0.0000
layer2.1.bn2.bias	32	0.0000
layer2.2.conv1.weight	949	0.8970
layer2.2.bn1.weight	32	0.0000
layer2.2.bn1.bias	32	0.0000
layer2.2.conv2.weight	922	0.9000
layer2.2.bn2.weight	32	0.0000
layer2.2.bn2.bias	32	0.0000
layer3.0.conv1.weight	1849	0.8997
layer3.0.bn1.weight	64	0.0000
layer3.0.bn1.bias	64	0.0000
layer3.0.conv2.weight	3653	0.9009
layer3.0.bn2.weight	64	0.0000
layer3.0.bn2.bias	64	0.0000
layer3.0.shortcut.0.weight	2048	0.0000
layer3.0.shortcut.1.weight	64	0.0000
layer3.0.shortcut.1.bias	64	0.0000
layer3.1.conv1.weight	3745	0.8984
layer3.1.bn1.weight	64	0.0000
layer3.1.bn1.bias	64	0.0000
layer3.1.conv2.weight	3634	0.9014
layer3.1.bn2.weight	64	0.0000
layer3.1.bn2.bias	64	0.0000

layer3.2.conv1.weight	3686	0.9000
layer3.2.bn1.weight	64	0.0000
layer3.2.bn1.bias	64	0.0000
layer3.2.conv2.weight	3660	0.9007
layer3.2.bn2.weight	64	0.0000
layer3.2.bn2.bias	64	0.0000
linear.weight	61	0.9047
linear.bias	0	1.0000

TOTAL NETWORK SPARSITY	30995	0.8862
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--- Sparsity Statistics: SGD | 95% Target ---

Layer Name	Active Weights	Actual Sparsity
conv1.weight	24	0.9444
bn1.weight	16	0.0000
bn1.bias	16	0.0000
layer1.0.conv1.weight	132	0.9427
layer1.0.bn1.weight	16	0.0000
layer1.0.bn1.bias	16	0.0000
layer1.0.conv2.weight	121	0.9475
layer1.0.bn2.weight	16	0.0000
layer1.0.bn2.bias	16	0.0000
layer1.1.conv1.weight	117	0.9492
layer1.1.bn1.weight	16	0.0000
layer1.1.bn1.bias	16	0.0000
layer1.1.conv2.weight	136	0.9410
layer1.1.bn2.weight	16	0.0000
layer1.1.bn2.bias	16	0.0000
layer1.2.conv1.weight	129	0.9440
layer1.2.bn1.weight	16	0.0000
layer1.2.bn1.bias	16	0.0000
layer1.2.conv2.weight	123	0.9466
layer1.2.bn2.weight	16	0.0000
layer1.2.bn2.bias	16	0.0000
layer2.0.conv1.weight	246	0.9466
layer2.0.bn1.weight	32	0.0000
layer2.0.bn1.bias	32	0.0000
layer2.0.conv2.weight	494	0.9464
layer2.0.bn2.weight	32	0.0000
layer2.0.bn2.bias	32	0.0000
layer2.0.shortcut.0.weight	512	0.0000

layer2.0.shortcut.1.weight	32	0.0000
layer2.0.shortcut.1.bias	32	0.0000
layer2.1.conv1.weight	455	0.9506
layer2.1.bn1.weight	32	0.0000
layer2.1.bn1.bias	32	0.0000
layer2.1.conv2.weight	461	0.9500
layer2.1.bn2.weight	32	0.0000
layer2.1.bn2.bias	32	0.0000
layer2.2.conv1.weight	456	0.9505
layer2.2.bn1.weight	32	0.0000
layer2.2.bn1.bias	32	0.0000
layer2.2.conv2.weight	454	0.9507
layer2.2.bn2.weight	32	0.0000
layer2.2.bn2.bias	32	0.0000
layer3.0.conv1.weight	919	0.9501
layer3.0.bn1.weight	64	0.0000
layer3.0.bn1.bias	64	0.0000
layer3.0.conv2.weight	1834	0.9502
layer3.0.bn2.weight	64	0.0000
layer3.0.bn2.bias	64	0.0000
layer3.0.shortcut.0.weight	2048	0.0000
layer3.0.shortcut.1.weight	64	0.0000
layer3.0.shortcut.1.bias	64	0.0000
layer3.1.conv1.weight	1850	0.9498
layer3.1.bn1.weight	64	0.0000
layer3.1.bn1.bias	64	0.0000
layer3.1.conv2.weight	1850	0.9498
layer3.1.bn2.weight	64	0.0000
layer3.1.bn2.bias	64	0.0000
layer3.2.conv1.weight	1870	0.9493
layer3.2.bn1.weight	64	0.0000
layer3.2.bn1.bias	64	0.0000
layer3.2.conv2.weight	1879	0.9490
layer3.2.bn2.weight	64	0.0000
layer3.2.bn2.bias	64	0.0000
linear.weight	28	0.9563
linear.bias	0	1.0000

TOTAL NETWORK SPARSITY	17706	0.9350

--- Sparsity Statistics: SGD | 98% Target ---

Layer Name	Active Weights	Actual Sparsity
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conv1.weight	11	0.9745
bn1.weight	16	0.0000
bn1.bias	16	0.0000
layer1.0.conv1.weight	28	0.9878
layer1.0.bn1.weight	16	0.0000
layer1.0.bn1.bias	16	0.0000
layer1.0.conv2.weight	44	0.9809
layer1.0.bn2.weight	16	0.0000
layer1.0.bn2.bias	16	0.0000
layer1.1.conv1.weight	39	0.9831
layer1.1.bn1.weight	16	0.0000
layer1.1.bn1.bias	16	0.0000
layer1.1.conv2.weight	49	0.9787
layer1.1.bn2.weight	16	0.0000
layer1.1.bn2.bias	16	0.0000
layer1.2.conv1.weight	47	0.9796
layer1.2.bn1.weight	16	0.0000
layer1.2.bn1.bias	16	0.0000
layer1.2.conv2.weight	51	0.9779
layer1.2.bn2.weight	16	0.0000
layer1.2.bn2.bias	16	0.0000
layer2.0.conv1.weight	98	0.9787
layer2.0.bn1.weight	32	0.0000
layer2.0.bn1.bias	32	0.0000
layer2.0.conv2.weight	183	0.9801
layer2.0.bn2.weight	32	0.0000
layer2.0.bn2.bias	32	0.0000
layer2.0.shortcut.0.weight	512	0.0000
layer2.0.shortcut.1.weight	32	0.0000
layer2.0.shortcut.1.bias	32	0.0000
layer2.1.conv1.weight	195	0.9788
layer2.1.bn1.weight	32	0.0000
layer2.1.bn1.bias	32	0.0000
layer2.1.conv2.weight	193	0.9791
layer2.1.bn2.weight	32	0.0000
layer2.1.bn2.bias	32	0.0000
layer2.2.conv1.weight	187	0.9797
layer2.2.bn1.weight	32	0.0000
layer2.2.bn1.bias	32	0.0000
layer2.2.conv2.weight	193	0.9791
layer2.2.bn2.weight	32	0.0000
layer2.2.bn2.bias	32	0.0000

layer3.0.conv1.weight	361	0.9804
layer3.0.bn1.weight	64	0.0000
layer3.0.bn1.bias	64	0.0000
layer3.0.conv2.weight	759	0.9794
layer3.0.bn2.weight	64	0.0000
layer3.0.bn2.bias	64	0.0000
layer3.0.shortcut.0.weight	2048	0.0000
layer3.0.shortcut.1.weight	64	0.0000
layer3.0.shortcut.1.bias	64	0.0000
layer3.1.conv1.weight	740	0.9799
layer3.1.bn1.weight	64	0.0000
layer3.1.bn1.bias	64	0.0000
layer3.1.conv2.weight	698	0.9811
layer3.1.bn2.weight	64	0.0000
layer3.1.bn2.bias	64	0.0000
layer3.2.conv1.weight	754	0.9795
layer3.2.bn1.weight	64	0.0000
layer3.2.bn1.bias	64	0.0000
layer3.2.conv2.weight	721	0.9804
layer3.2.bn2.weight	64	0.0000
layer3.2.bn2.bias	64	0.0000
linear.weight	15	0.9766
linear.bias	1	0.9000

TOTAL NETWORK SPARSITY	9495	0.9652

--- Sparsity Statistics: ADAM | 90% Target ---

Layer Name	Active Weights	Actual Sparsity

conv1.weight	48	0.8889
bn1.weight	16	0.0000
bn1.bias	16	0.0000
layer1.0.conv1.weight	245	0.8937
layer1.0.bn1.weight	16	0.0000
layer1.0.bn1.bias	16	0.0000
layer1.0.conv2.weight	230	0.9002
layer1.0.bn2.weight	16	0.0000
layer1.0.bn2.bias	16	0.0000
layer1.1.conv1.weight	235	0.8980
layer1.1.bn1.weight	16	0.0000
layer1.1.bn1.bias	16	0.0000
layer1.1.conv2.weight	248	0.8924

layer1.1.bn2.weight	16	0.0000
layer1.1.bn2.bias	16	0.0000
layer1.2.conv1.weight	220	0.9045
layer1.2.bn1.weight	16	0.0000
layer1.2.bn1.bias	16	0.0000
layer1.2.conv2.weight	201	0.9128
layer1.2.bn2.weight	16	0.0000
layer1.2.bn2.bias	16	0.0000
layer2.0.conv1.weight	471	0.8978
layer2.0.bn1.weight	32	0.0000
layer2.0.bn1.bias	32	0.0000
layer2.0.conv2.weight	935	0.8985
layer2.0.bn2.weight	32	0.0000
layer2.0.bn2.bias	32	0.0000
layer2.0.shortcut.0.weight	512	0.0000
layer2.0.shortcut.1.weight	32	0.0000
layer2.0.shortcut.1.bias	32	0.0000
layer2.1.conv1.weight	920	0.9002
layer2.1.bn1.weight	32	0.0000
layer2.1.bn1.bias	32	0.0000
layer2.1.conv2.weight	896	0.9028
layer2.1.bn2.weight	32	0.0000
layer2.1.bn2.bias	32	0.0000
layer2.2.conv1.weight	922	0.9000
layer2.2.bn1.weight	32	0.0000
layer2.2.bn1.bias	32	0.0000
layer2.2.conv2.weight	936	0.8984
layer2.2.bn2.weight	32	0.0000
layer2.2.bn2.bias	32	0.0000
layer3.0.conv1.weight	1869	0.8986
layer3.0.bn1.weight	64	0.0000
layer3.0.bn1.bias	64	0.0000
layer3.0.conv2.weight	3745	0.8984
layer3.0.bn2.weight	64	0.0000
layer3.0.bn2.bias	64	0.0000
layer3.0.shortcut.0.weight	2048	0.0000
layer3.0.shortcut.1.weight	64	0.0000
layer3.0.shortcut.1.bias	64	0.0000
layer3.1.conv1.weight	3678	0.9002
layer3.1.bn1.weight	64	0.0000
layer3.1.bn1.bias	64	0.0000
layer3.1.conv2.weight	3667	0.9005
layer3.1.bn2.weight	64	0.0000

layer3.1.bn2.bias	64	0.0000
layer3.2.conv1.weight	3715	0.8992
layer3.2.bn1.weight	64	0.0000
layer3.2.bn1.bias	64	0.0000
layer3.2.conv2.weight	3771	0.8977
layer3.2.bn2.weight	64	0.0000
layer3.2.bn2.bias	64	0.0000
linear.weight	66	0.8969
linear.bias	3	0.7000

TOTAL NETWORK SPARSITY	31149	0.8857
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--- Sparsity Statistics: ADAM | 95% Target ---

Layer Name	Active Weights	Actual Sparsity
conv1.weight	26	0.9398
bn1.weight	16	0.0000
bn1.bias	16	0.0000
layer1.0.conv1.weight	102	0.9557
layer1.0.bn1.weight	16	0.0000
layer1.0.bn1.bias	16	0.0000
layer1.0.conv2.weight	132	0.9427
layer1.0.bn2.weight	16	0.0000
layer1.0.bn2.bias	16	0.0000
layer1.1.conv1.weight	104	0.9549
layer1.1.bn1.weight	16	0.0000
layer1.1.bn1.bias	16	0.0000
layer1.1.conv2.weight	130	0.9436
layer1.1.bn2.weight	16	0.0000
layer1.1.bn2.bias	16	0.0000
layer1.2.conv1.weight	114	0.9505
layer1.2.bn1.weight	16	0.0000
layer1.2.bn1.bias	16	0.0000
layer1.2.conv2.weight	109	0.9527
layer1.2.bn2.weight	16	0.0000
layer1.2.bn2.bias	16	0.0000
layer2.0.conv1.weight	261	0.9434
layer2.0.bn1.weight	32	0.0000
layer2.0.bn1.bias	32	0.0000
layer2.0.conv2.weight	466	0.9494
layer2.0.bn2.weight	32	0.0000
layer2.0.bn2.bias	32	0.0000

layer2.0.shortcut.0.weight	512	0.0000
layer2.0.shortcut.1.weight	32	0.0000
layer2.0.shortcut.1.bias	32	0.0000
layer2.1.conv1.weight	450	0.9512
layer2.1.bn1.weight	32	0.0000
layer2.1.bn1.bias	32	0.0000
layer2.1.conv2.weight	518	0.9438
layer2.1.bn2.weight	32	0.0000
layer2.1.bn2.bias	32	0.0000
layer2.2.conv1.weight	426	0.9538
layer2.2.bn1.weight	32	0.0000
layer2.2.bn1.bias	32	0.0000
layer2.2.conv2.weight	477	0.9482
layer2.2.bn2.weight	32	0.0000
layer2.2.bn2.bias	32	0.0000
layer3.0.conv1.weight	876	0.9525
layer3.0.bn1.weight	64	0.0000
layer3.0.bn1.bias	64	0.0000
layer3.0.conv2.weight	1835	0.9502
layer3.0.bn2.weight	64	0.0000
layer3.0.bn2.bias	64	0.0000
layer3.0.shortcut.0.weight	2048	0.0000
layer3.0.shortcut.1.weight	64	0.0000
layer3.0.shortcut.1.bias	64	0.0000
layer3.1.conv1.weight	1844	0.9500
layer3.1.bn1.weight	64	0.0000
layer3.1.bn1.bias	64	0.0000
layer3.1.conv2.weight	1875	0.9491
layer3.1.bn2.weight	64	0.0000
layer3.1.bn2.bias	64	0.0000
layer3.2.conv1.weight	1804	0.9511
layer3.2.bn1.weight	64	0.0000
layer3.2.bn1.bias	64	0.0000
layer3.2.conv2.weight	1787	0.9515
layer3.2.bn2.weight	64	0.0000
layer3.2.bn2.bias	64	0.0000
linear.weight	23	0.9641
linear.bias	2	0.8000

TOTAL NETWORK SPARSITY	17489	0.9358

--- Sparsity Statistics: ADAM | 98% Target ---

Layer Name	Active Weights	Actual Sparsity

conv1.weight	11	0.9745
bn1.weight	16	0.0000
bn1.bias	16	0.0000
layer1.0.conv1.weight	64	0.9722
layer1.0.bn1.weight	16	0.0000
layer1.0.bn1.bias	16	0.0000
layer1.0.conv2.weight	62	0.9731
layer1.0.bn2.weight	16	0.0000
layer1.0.bn2.bias	16	0.0000
layer1.1.conv1.weight	44	0.9809
layer1.1.bn1.weight	16	0.0000
layer1.1.bn1.bias	16	0.0000
layer1.1.conv2.weight	52	0.9774
layer1.1.bn2.weight	16	0.0000
layer1.1.bn2.bias	16	0.0000
layer1.2.conv1.weight	44	0.9809
layer1.2.bn1.weight	16	0.0000
layer1.2.bn1.bias	16	0.0000
layer1.2.conv2.weight	37	0.9839
layer1.2.bn2.weight	16	0.0000
layer1.2.bn2.bias	16	0.0000
layer2.0.conv1.weight	91	0.9803
layer2.0.bn1.weight	32	0.0000
layer2.0.bn1.bias	32	0.0000
layer2.0.conv2.weight	180	0.9805
layer2.0.bn2.weight	32	0.0000
layer2.0.bn2.bias	32	0.0000
layer2.0.shortcut.0.weight	512	0.0000
layer2.0.shortcut.1.weight	32	0.0000
layer2.0.shortcut.1.bias	32	0.0000
layer2.1.conv1.weight	176	0.9809
layer2.1.bn1.weight	32	0.0000
layer2.1.bn1.bias	32	0.0000
layer2.1.conv2.weight	181	0.9804
layer2.1.bn2.weight	32	0.0000
layer2.1.bn2.bias	32	0.0000
layer2.2.conv1.weight	195	0.9788
layer2.2.bn1.weight	32	0.0000
layer2.2.bn1.bias	32	0.0000
layer2.2.conv2.weight	177	0.9808
layer2.2.bn2.weight	32	0.0000

layer2.2.bn2.bias	32	0.0000
layer3.0.conv1.weight	334	0.9819
layer3.0.bn1.weight	64	0.0000
layer3.0.bn1.bias	64	0.0000
layer3.0.conv2.weight	783	0.9788
layer3.0.bn2.weight	64	0.0000
layer3.0.bn2.bias	64	0.0000
layer3.0.shortcut.0.weight	2048	0.0000
layer3.0.shortcut.1.weight	64	0.0000
layer3.0.shortcut.1.bias	64	0.0000
layer3.1.conv1.weight	753	0.9796
layer3.1.bn1.weight	64	0.0000
layer3.1.bn1.bias	64	0.0000
layer3.1.conv2.weight	744	0.9798
layer3.1.bn2.weight	64	0.0000
layer3.1.bn2.bias	64	0.0000
layer3.2.conv1.weight	729	0.9802
layer3.2.bn1.weight	64	0.0000
layer3.2.bn1.bias	64	0.0000
layer3.2.conv2.weight	693	0.9812
layer3.2.bn2.weight	64	0.0000
layer3.2.bn2.bias	64	0.0000
linear.weight	19	0.9703
linear.bias	1	0.9000

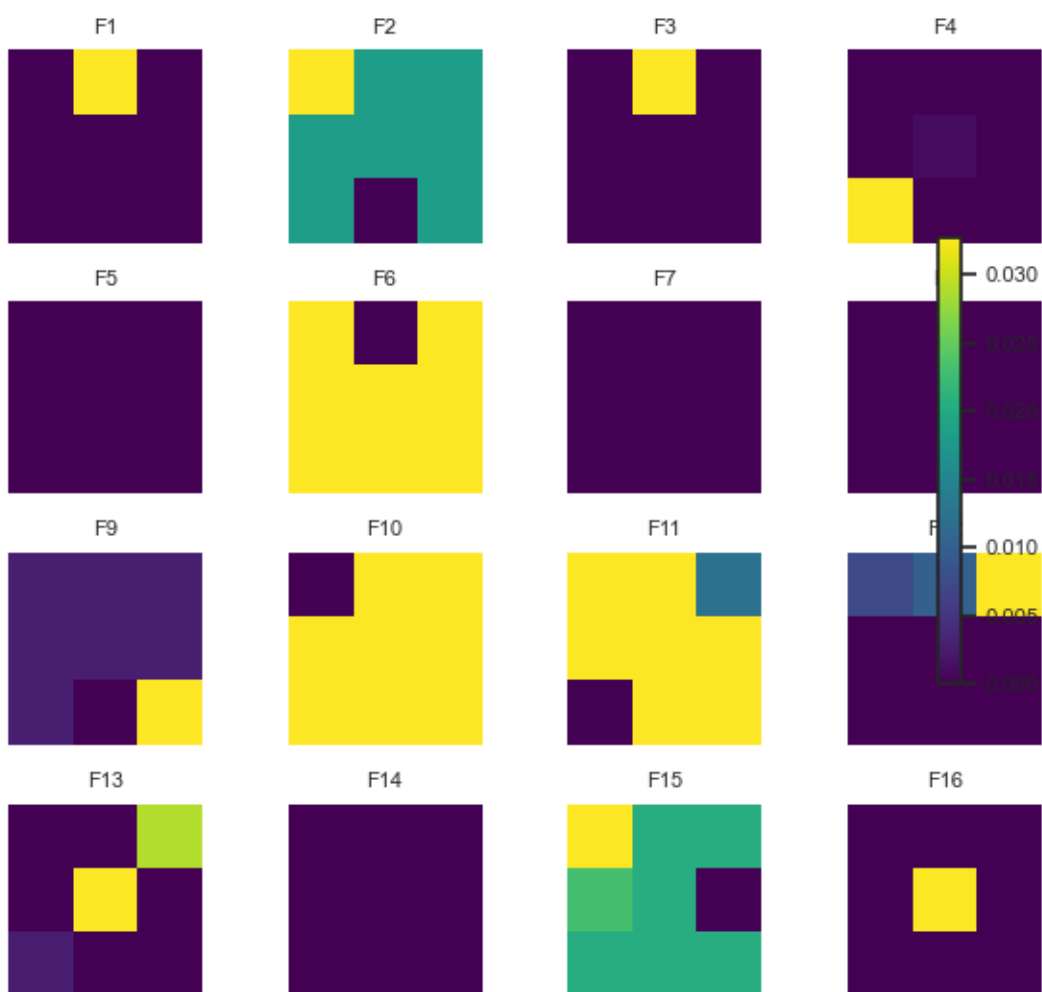
TOTAL NETWORK SPARSITY	9498	0.9651

Visualize kernels

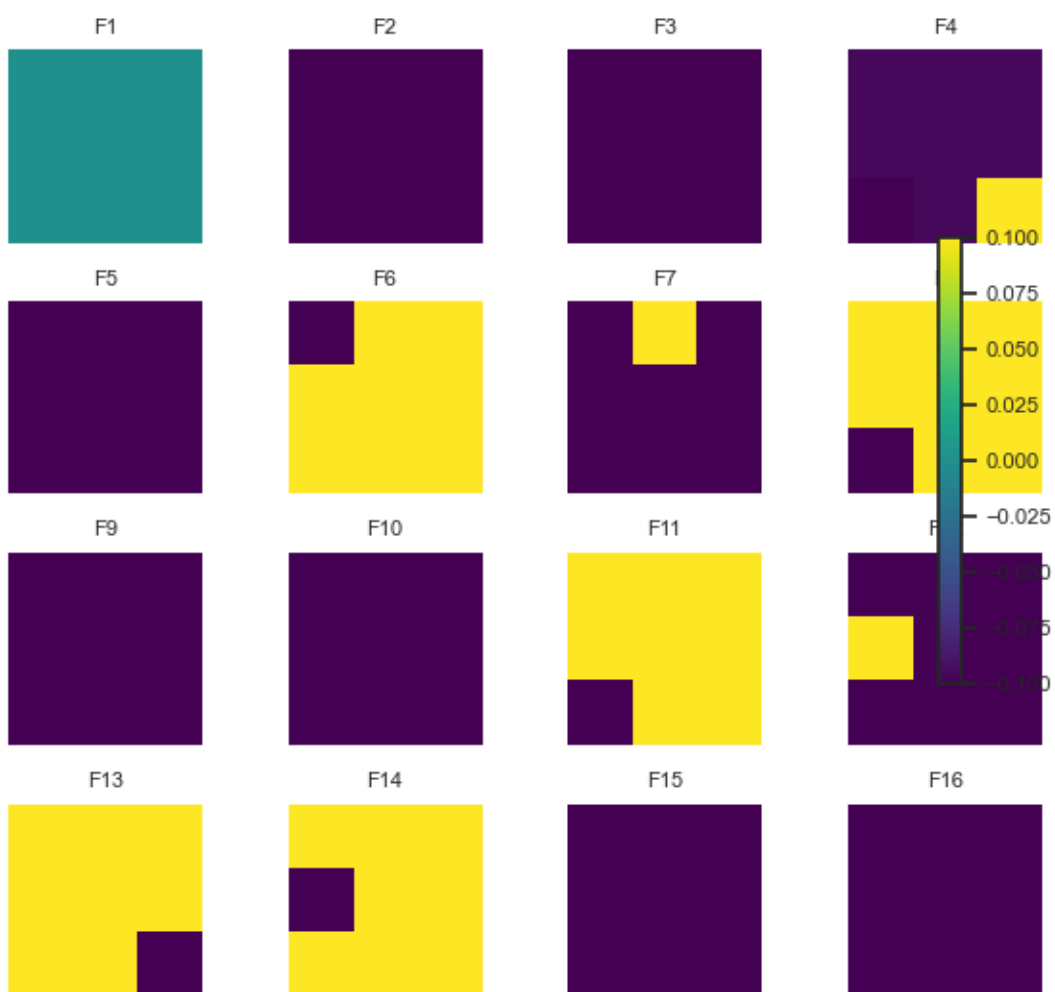
```
for opt in optimizers:
    for s in sparsities:
        visualize_kernels(results, opt=opt, s=s)
```

```
/var/folders/5m/q3t_7tx111750b5hqf72gt3r0000gn/T/ipykernel_38890/507913117.py:31: UserWarning:
plt.tight_layout()
```

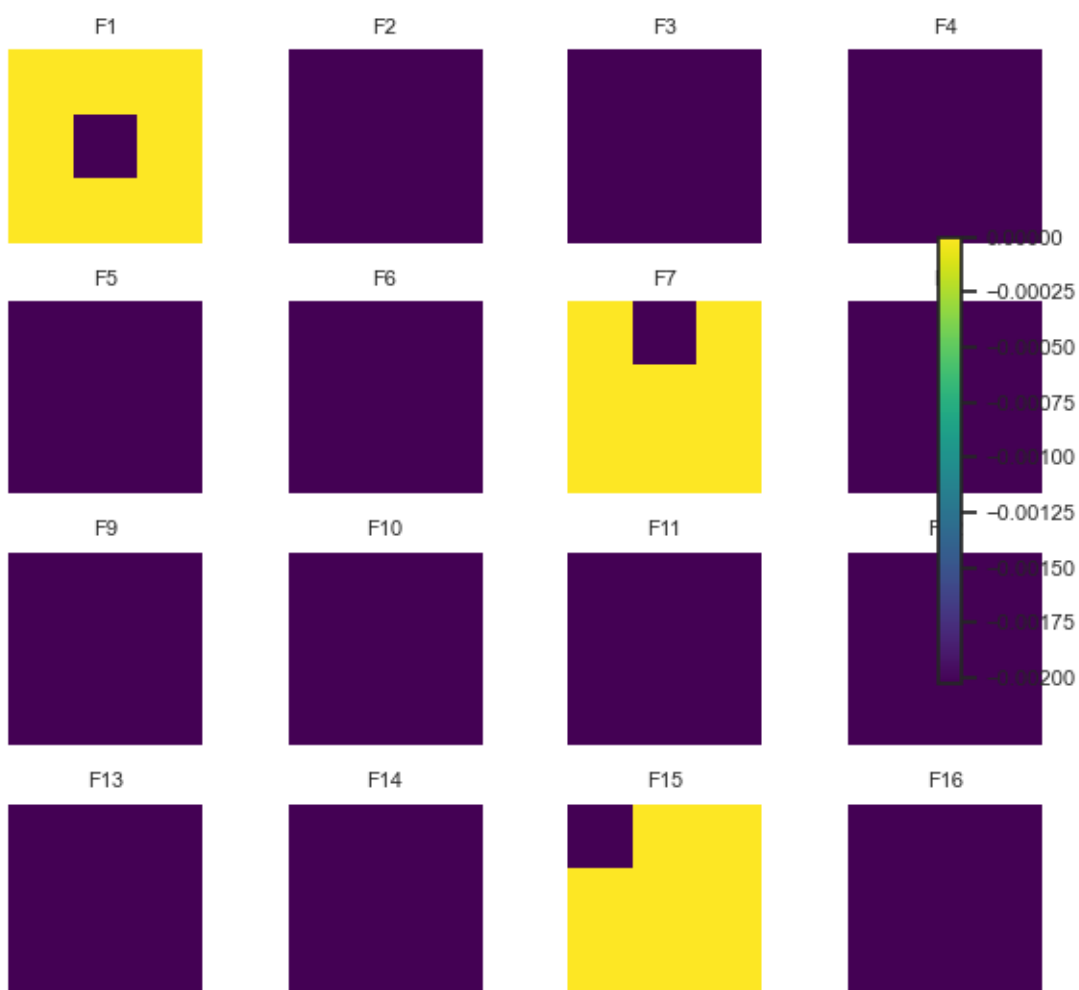
Kernels (layer1.0.conv1.weight) SGD | 90% Sparsity



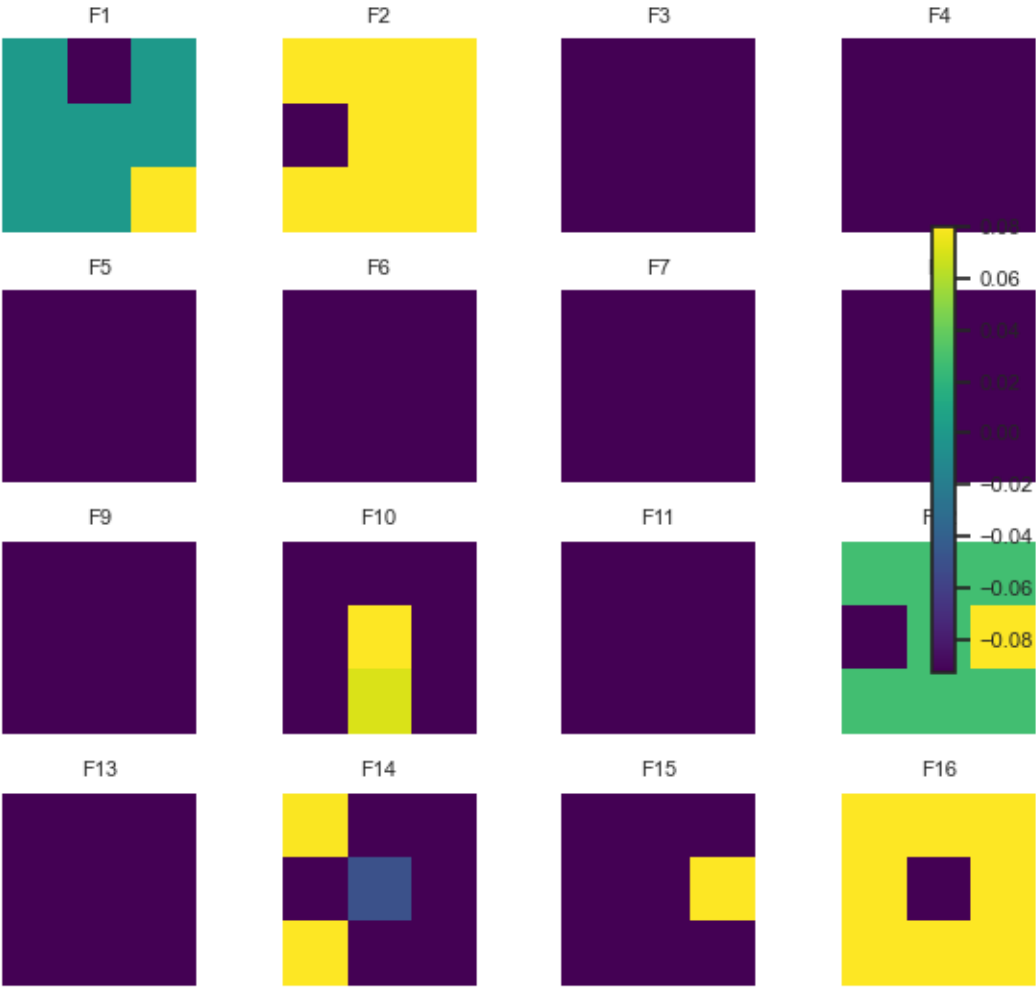
Kernels (layer1.0.conv1.weight) SGD | 95% Sparsity



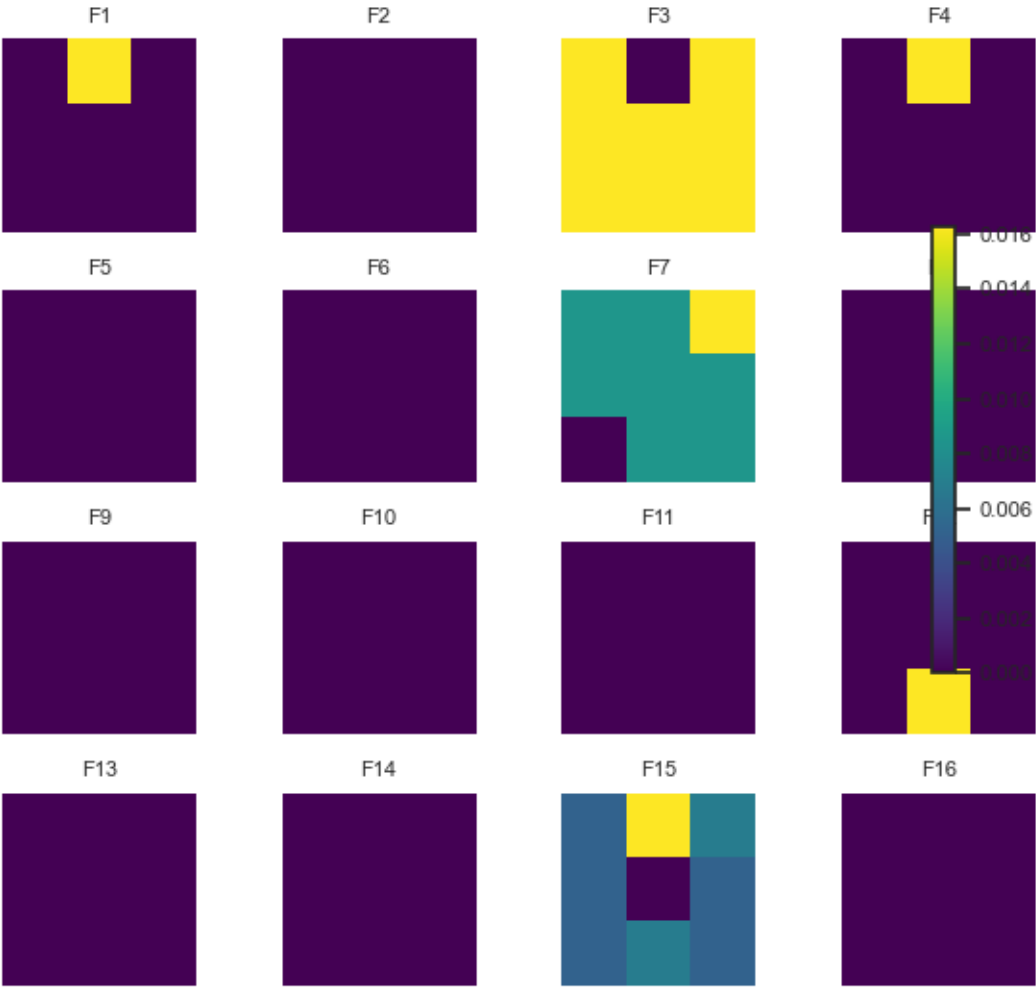
Kernels (layer1.0.conv1.weight) SGD | 98% Sparsity



Kernels (layer1.0.conv1.weight)
ADAM | 90% Sparsity



Kernels (layer1.0.conv1.weight)
ADAM | 95% Sparsity



Kernels (layer1.0.conv1.weight)
ADAM | 98% Sparsity

